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Roll No.....

Degree: B.Tech. Semester: III - Course work

MID-SEMESTER EXAMINATION, September, 2026

Course Title: *Mathematics for Communication and Signal Processing*

Course Code: ECMTC301

Time: 1:30 Hours **Max. Marks: 20**

Note: Attempt all questions in given order only. Missing data/ information (if any), maybe suitably assumed and mentioned in the answer.

Q. No.	Question	Marks	CO
1*a*	Prove that the set $H = \{A \in M_3(\mathbb{Q}) : \text{trace}(A^T) = 0\}$ forms a subgroup of the group $(M_3(\mathbb{R}), +)$ of all 3×3 real matrices under matrix addition.	2	CO1
1*b*	Let $$G = \left\{ \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} : a \in \mathbb{Z}_6 \right\}$$ be a group under matrix multiplication. Determine the order of the group G. Also, Find the order of each element of G.	2	CO1
2*a*	Define a cyclic group. Determine whether the group $U(16)$ is cyclic? Justify your answer.	2	CO1
2*b*	Let $f : \mathbb{Z}_{12} \to \mathbb{Z}_{12}$ be defined by $f(x) = 3x \pmod{12}$. Prove that f is a group homomorphism. Hence, find $\ker(f)$.	2	CO1
3*a*	Consider the ring $\mathbb{Z}_{10} = \mathbb{Z}/10\mathbb{Z}$. (a) Check whether $\mathbb{Z}_{10}$ has any zero divisors. (b) Is $\mathbb{Z}_{10}$ a field? Justify.	2	CO1
3*b*	Whether the set $W = \{(x, y, z, w) \in \mathbb{R}^4 \mid 4x - 5y = 7z + w\}$ forms a subspace of $\mathbb{R}^4$? Justify your answer.	2	CO2
4a	Investigate the linear independence or dependence of the set of vectors $x_1 = (1, 0, 2)$, $x_2 = (0, 1, -1)$, $x_3 = (2, 1, 3)$, $x_4 = (3, 1, 2)$ in $\mathbb{R}^3$.	2	CO2
4*b*	Consider the vector space $\mathbb{P}_2$ of polynomials of degree at most 2. Is the set $\{2 + x, x + x^2, 2 + x^2\}$ forms a spanning set for the vector space $\mathbb{P}_2$? Justify your answer.	2	CO2
5*a*	Using Rank Nullity Theorem, show that the linear transformation $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^4$ given by $T(x, y, z) = (x - y, y - z, z - x, 0)$ is not bijective.	2	CO2
5*b*	For the vector field $\mathbf{F}(x, y, z) = (x^2y)\mathbf{i} + (yz^2)\mathbf{j} + (xz)\mathbf{k}$: (i) Compute the curl; $\nabla \times \mathbf{F}$. (ii) Compute the divergence of the curl, $\nabla \cdot (\nabla \times \mathbf{F})$.	2	CO3